

genesis of the RA joint lesions, logic suggests the radical approach of B-cell ablation with monoclonal anti-CD20 as used in the treatment of B-cell leukemia. Results to date are provocatively encouraging.

Manipulation of regulatory mediators

We can correct some spontaneous models of autoimmune disease by injection of cytokines: IL-1 cures the diabetes of NOD mice; TNF prevents the onset of SLE symptoms in NZB×W hybrids; and transforming growth factor- β 1 (TGF β 1) is known to protect against collagen arthritis and relapsing EAE. We have already reminded ourselves of the maintenance doses of steroids to restore the defective adrenal feedback control on leukocytes in RA.

If we now take it as almost gospel that the Th1 subset is pathogenic in solid organ-specific disease, attempts to switch the phenotype to Th2 should be beneficial. On the assumption that IL-4 or IL-10 can effect this switch, then treatment of an autoimmune individual with these cytokines should defuse the Th1 cells, as was indeed observed in an EAE model. The exacerbation of Th1-mediated disease by IL-12 raises the possibility that the highly avid dimer of one of the IL-2 receptor chains might be a potent inhibitor. However, with the exception of steroids in RA, there is quite a gap between therapy in experimental animals and its application to human disease. Our earlier suggestion of transfection of the target organ with a protective cytokine like TGF β (figure 20.22 (8)) may prove to be a good bet (visions of firing a biolistic gun at islets pretransplantation?).

Idiotype control with antibody

The powerful immunosuppressive action of anti-idiotypic antibodies has led to much rumination on the feasibility of controlling autoantibody production by provoking appropriate interactions within the immune network. We have focused previously on the intimate network interactions between hormone receptors, hormones and their respective antibodies (cf. p. 209) and it might be that the autoimmune disorders involving these receptors are especially amenable to idiotype control. There is a growing realization that, in general, more fundamental suppression can be achieved by utilizing the internal elements of the idiotype network rather than anti-idiotypic reagents raised in other species. Thus, xenogeneic anti-idiotypes have only won transient and partial improvement in the spontaneous thyroiditis of the Buffalo rat and the autoimmune lupus of NZB×W mice, presumably due

to compensation by idiotype-negative clones. On the other hand, much more profound changes have been achieved by treatment with monoclonal autoantibodies (idiotypes) derived from the autoimmune strain in question. Thus, prior administration without adjuvant of two peptides taken from the CDRs of a pathogenic murine 16/6 Id⁺ anti-DNA monoclonal was reported to inhibit the induction of SLE by immunization with the intact monoclonal in CFA.

Curiously, **intravenous injection of Ig pooled from many normal donors** is of positive benefit in a number of autoimmune blood diseases, recurrent abortions associated with cardiolipin antibodies, juvenile dermatomyositis and patients with autoantibodies to procoagulant factor VIII. The latter has been studied in some detail and the inhibitory effects of F(ab')₂ fractions from the normal Ig pool suggest that we are dealing with anti-idiotypic reactions; it is as though the normal pool was re-establishing a properly controlled network. These are intriguing observations which deserve serious consideration, although to some extent their use is marred by expense.

Vaccination with T-cell idiotypes

It is possible to protect animals against the induction of experimental allergic encephalomyelitis by immunization with an attenuated T-cell clone specific for myelin basic protein (MBP). This must be mediated by the induction of suppressor T-cells specific for the effector cell receptor idiotype. Confirmation has come from experiments showing that the encephalitis can be prevented if mice are first immunized with a synthetic peptide from the V β chain of the encephalitogenic clone; this procedure generates CD8 T-cells specific for the receptor peptide presented by class I MHC, and which transfer protection against induction of encephalitis.

This gambit (figure 20.22 (5)) has now been played in human disease. A TCR peptide vaccine embodying the V β 5.2 sequence expressed in MS plaques and on T-cells specific for MBP was used to treat MS patients in a double-blind trial (10 μ g weekly for 4 weeks and then monthly for 10 months). Lack of response to vaccination was associated with increased response to MBP and clinical progression, but successful vaccination boosted the frequency of TCR peptide-specific T-cells, reduced the frequency of MBP-specific cells and prevented clinical progression without side-effects. The reactive cells were predominantly Th2-like and directly inhibited MBP-specific Th1 responses, primarily through release of IL-10 and probably through an anti-idiotypic regulatory network. Although many of the

T-cells in the lesions would not belong to the V β 5.2 family, they could be switched off by organ-related bystander tolerance (see figure 20.22).

T-cell vaccination has been used to protect against the spontaneous development of diabetes in NOD mice and the production of arthritis following sensitization with type II collagen. It has also proved possible to switch off Freund adjuvant-induced arthritis with an attenuated clone of T-cells generated in response to the 65 kDa mycobacterial heat-shock protein. This adjuvant model has been looked at in depth (I. Cohen). Perversely, the earliest T-cell responses preceding the adjuvant-induced arthritis were antigen-specific suppression and anti-idiotypic reactivity; responses to the antigen itself emerged a few days before the appearance of clinical arthritis. T-cell vaccination accelerated the kinetics of the antigen response, abolished antigen-specific suppression, activated anti-idiotypic T-cells and inhibited arthritis. The extremely rapid appearance of anti-idiotypic and antigen-specific suppressors so soon after immunization with the 65 kDa heat-shock protein again strongly suggests a pre-existing network linked to epitopes on this antigen as envisaged in the 'immunological homunculus' concept (cf. p. 208). If malfunctioning of the network produces autoimmune disease, vaccination with T-cell receptor epitopes would represent a logical attempt to re-establish natural control.

Manipulation by antigen

The object is to present the offending antigen in sufficient concentration and in the form which will turn off an ongoing autoimmune response. Since T-cells have been accorded such a pivotal role, it is natural to devise the strategy in terms of T-cell epitopes rather than whole antigen, obviously a far more practical proposition because this reduces the problem to dealing with relatively short peptides. One strategy is to design high affinity peptide analogs that will bind obstinately to the appropriate MHC molecule and antagonize the response to autoantigen (figure 20.22 (4)). Since we express several different MHC molecules, this should not impair microbial defenses unduly. However, we are now talking of patients not mice and this could involve repeated very high doses of peptide, although, much in their favor, peptides are well defined chemically and *relatively* cheap to produce. Antigen-specific suppression of T-cells (figure 20.22 (3)) would be advantageous in this respect, and giving the peptide under an umbrella of anti-CD4 or using partial agonists (cf. figure 9.8) could be feasible. Injection of an MBP peptide, particularly as a palmitoylated deriva-

tive inserted in liposomes, can block EAE and an hsp60 peptide can prevent the onset of diabetes in the NOD mouse (cf. p. 434). Awareness of the therapeutic benefit of injected insulin in the NOD model has fostered a large-scale trial in the human disease and considerable clinical improvement has been achieved in patients with exacerbating-remitting MS given Cop1, a random copolymer of alanine, glutamic acid, lysine and tyrosine meant to simulate MBP. Many thousands of patients have been injected with Cop1 subcutaneously every day, thereby reducing the risk of relapse from 1.5 per year at onset to less than 1 in every 5 years without an increase in neurological disability. The copolymer competes with MBP and other brain antigens for binding to the MHC and to the cognate T-cell receptors and induces T-suppressors which, it may be said, are also generated when the antigen is fed.

We have already noted that, because the mucosal surface of the gut is exposed to a horde of powerfully immunogenic microorganisms, and since enterocytes are especially vulnerable to damage by IFN γ and TNF, it has been important for the immune defenses of the gut to evolve mechanisms which deter Th1-type responses. This objective is attained by the stimulation of cells which release cytokines such as TGF β , IL-4 and IL-10 and suppress the unwanted responses. Thus feeding antigens should tolerize Th1 cells and this has proved to be a successful strategy for blocking EAE, the collagen II arthritis model and the development of diabetes in NOD mice. Accordingly, MS patients are now being fed MBP and RA patients type II chicken collagen.

The tolerogen can also be delivered by inhalation of peptide aerosols (figure 20.23), and this could be a very attractive way of generating antigen-specific T-cell suppression in many hypersensitivity states. Induction of anergy or active suppression may contribute to different extents. Intranasal peptides have been used successfully to block collagen-induced arthritis, EAE, spontaneous diabetes (NOD) and a mouse model of allergy to the house dust mite antigen Der P1. Significantly, treatment can be effective even *after* induction of disease (figure 20.23), although in established human disease this may be more difficult to achieve and might require supplementary therapy, such as anti-CD4, and preliminary reduction of primed T-cells with cyclosporin or steroids. There is no shortage of strings to pull.

Now this is really important. A single internal epitope of MBP can inhibit disease induced by the *mixture* of epitopes or antigens contained within whole myelin. In other words, a single epitope can induce

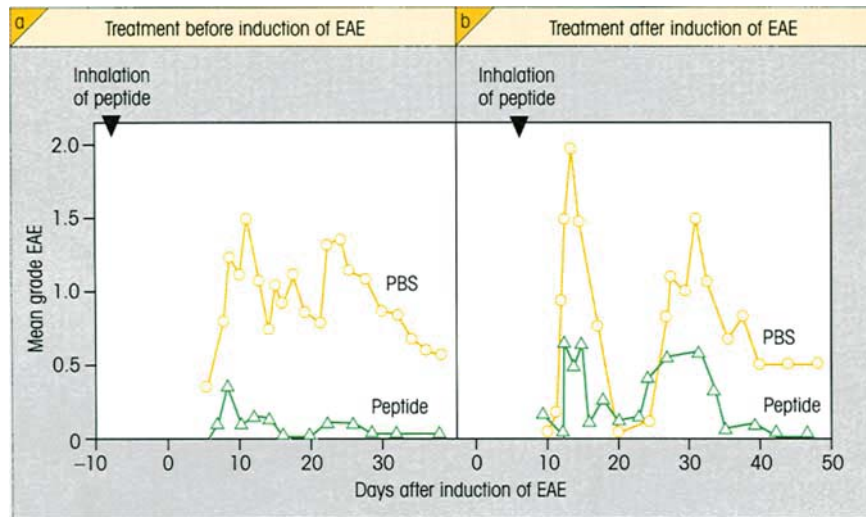


Figure 20.23. Influence of peptide inhalation on experimental autoimmune encephalomyelitis (EAE) induced with pig spinal cord in complete Freund's adjuvant. Aerosols of the peptides were inhaled (a) 8 days before and (b) 8 days after injection of the encephalitogen. A single dose can give long-lived protection which is extended indefinitely if the mice are thymectomized. Regulation is IL-10 dependent and both Th1 and Th2 can be tolerized. Administration of a single peptide T-cell epitope can in-

duce tolerance to other autoantigenic epitopes on the same protein (linked suppression) and to epitopes on different antigens within the nervous tissue used for immunization (bystander tolerance). PBS, phosphate-buffered saline; the peptide was an acetylated N-terminal 11-mer from myelin basic protein with lysine at position 4 substituted by alanine. (Data from Metzler B. & Wraith D.C. (1996) *Annals of the New York Academy of Science* **778**, 228, with permission of the publishers.)

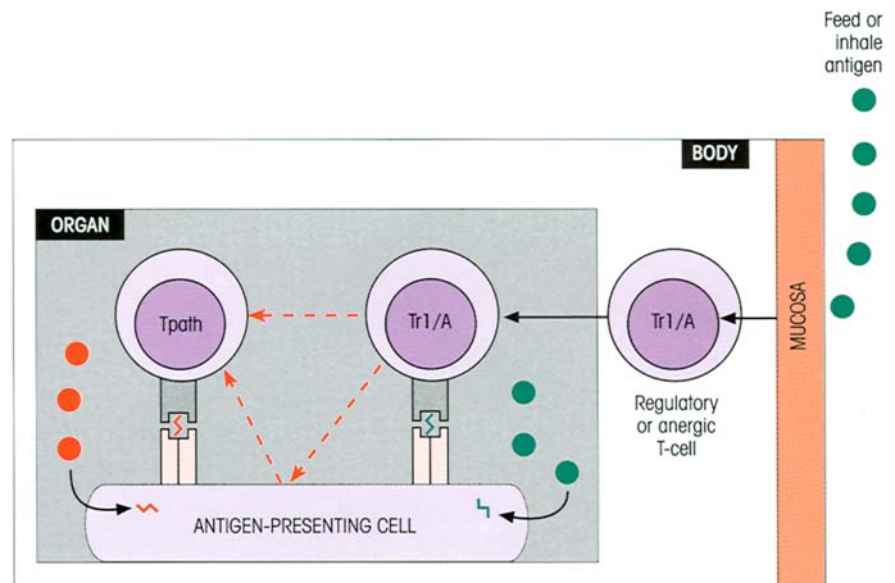


Figure 20.24. Organ-related bystander tolerance induced by feeding or inhaling an organ-related autoantigen. Induced regulatory or anergic tolerogen-specific T-cells (Tr1/A) enter the organ and inhibit pathogenic T-cells (Tpath) on the same antigen-presenting cell which processes both the tolerogen and the other organ-derived antigen recognized by the pathogenic cell. Regulators act by production of IL-10 and possibly TGF β , which downregulate the Th1 cells either directly or through an intermediate effect on the antigen-presenting cell.

suppression of the pathogenic T-cells specific for other epitopes on the same or other molecules provided that they are generated within the same organ or locality. We have referred to this already as **organ-related bystander tolerance**, a phenomenon best understood in terms of interactions on the same antigen-presenting cell between the regulatory cell, be it Th2 or anergic, recognizing the suppressor epitope and the patho-

genic Th1 cell recognizing a **separate epitope** processed from the same or another molecule in the same organ (figure 20.24).

Another potentially valuable approach for the future involves 'switching off' primed B-cells by presenting hapten linked to a thymus-independent carrier like the copolymer of D-glutamic acid and D-lysine (D-GL) or IgG of the same species, particularly when given

with high cortisone doses. This has certainly worked well in NZB×W hybrid mice where anti-DNA levels have been reduced using nucleosides as the haptens: we shall have to see whether human and mouse really are that different.

Another way to manipulate antigen is to remove it. For nucleosomal DNA this can be accomplished by injection of DNase. Sure enough, treatment with DNase I suppressed the manifestations of lupus nephritis in the NZB×W model. And last, back to the Holy Grail business, several groups are trying to evolve a strategy based upon the 'magic bullet', the essence of which is to fashion different types of cytotoxic weaponry by coupling bacterial toxins or lots of radioactivity to the antigen which selectively homes on to the lymphocytes bearing specific surface receptors. Something good has got to come out of all this!

Plasmapheresis

Plasma exchange to lower the rate of immune complex deposition in SLE provides only temporary benefit, although it may be of value in life-threatening cases of arteritis. Successful results have been obtained in Goodpasture's syndrome when the treatment has been applied in combination with antimetabolic drugs (figure 20.25), the rationale being an increased tendency for antigen-reactive cells to divide as the negative feedback effect of IgG is lowered following removal of plasma proteins.

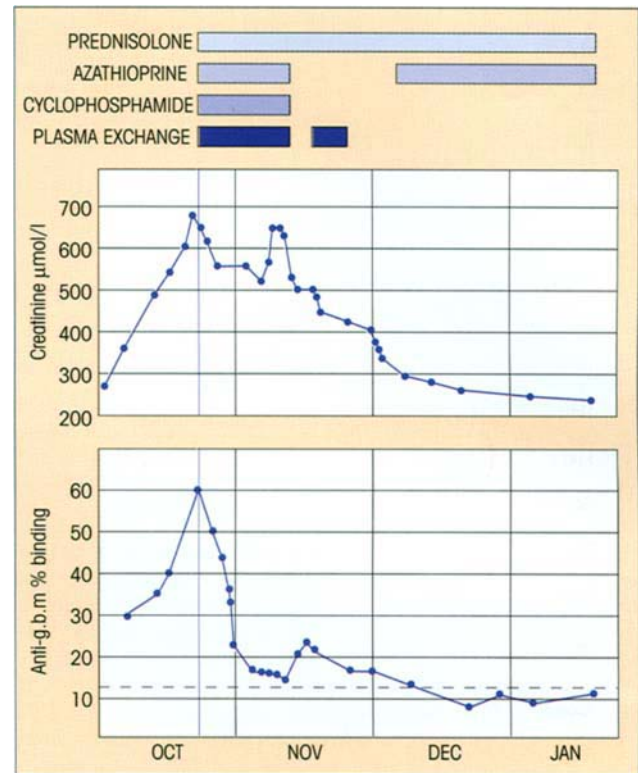


Figure 20.25. Treatment of a patient with anti-glomerular basement membrane (anti-g.b.m.)-induced nephritis with plasma exchange, steroids and immunosuppressive drugs. Kidney function is here monitored by the serum creatinine level. The treatment leads to loss of autoantibody (the dashed line represents the amount of g.b.m. antigen bound in the assay by normal serum) and restoration of kidney function. (Courtesy of Dr C.M. Lockwood.)

SUMMARY

Pathogenic effects of humoral autoantibody

- Direct pathogenic effects of human autoantibodies to blood, surface receptors and several other tissues are listed in table 20.2.
- Passive transfer of disease is seen in 'experiments of nature' in which transplacental passage of maternal IgG autoantibody produces a comparable but transient disorder in the fetus and neonate.
- Disease can also be mimicked in experimental animals by passive transfer of monoclonal autoantibodies.

Pathogenic effects of complexes with autoantigens

- Immune complexes, usually with bound complement, appear in the kidneys, skin and joints of patients with SLE, associated with lesions in the corresponding organs.
- The formation of high affinity, mutated IgG anti-

bodies to anatomically clustered antigens (e.g. nucleosome components) attests to T-cell control and antigen selection of the antibody response. These antigen clusters can appear as blebs on the surface of apoptotic cells.

- Spontaneous lupus has been observed in certain pure-bred animal strains and the importance of autoimmunity for pathogenesis is shown by the amelioration of symptoms whenever the immune response is suppressed.
- The IgG in RA shows defective galactosylation of the Fc sugars.
- Most patients with RA produce autoantibodies to IgG (rheumatoid factors) as a result of immunological hyperreactivity in the deeper layers of the synovium. The IgG rheumatoid factors self-associate to form complexes.

(continued)

Table 20.2. Direct pathogenic effects of humoral antibodies.

DISEASE	AUTOANTIGEN	LESION
Autoimmune hemolytic anemia	Red cell	Erythrocyte destruction
Lymphopenia (some cases)	Lymphocyte	Lymphocyte destruction
Idiopathic thrombocytopenic purpura	Platelet	Platelet destruction
Anti-phospholipid syndrome	Cardiolipin/ β 2-glycoprotein I complex	Recurrent thromboembolic phenomena
Male infertility (some cases)	Sperm	Agglutination of spermatozoa
Pernicious anemia	H ⁺ /K ⁺ -ATPase, gastrin receptor	Block acid production
Hashimoto's disease	Thyroid peroxidase surface antigen	Cytotoxic effect on thyroid cells in culture
Primary myxedema	TSH receptor	Blocking of thyroid cell
Graves' disease	TSH receptor	Stimulation of thyroid cell
Goodpasture's syndrome	Glomerular basement membrane	Complement-mediated damage to basement membrane
Myasthenia gravis	Acetylcholine receptor	Blocking and destruction of receptors
Lambert-Eaton syndrome	Presynaptic Ca channel	Neuromuscular defect
Acanthosis nigricans (type B) and ataxia telangiectasia with insulin resistance	Insulin receptor	Blocking of receptors
Atopic allergy (some cases)	β -Adrenergic receptors	Blocking of receptors
Congenital heart block	Ro/SS-A	Distort fetal cardiac membrane action potential
Celiac disease	Endomysium	Small intestinal inflammation

- These give rise to acute inflammation in the joint space and stimulate the synovial lining cells to grow as a malign **pannus** which **produces erosions in the underlying cartilage and bone** through the release of IL-1, IL-6, TNF, prostaglandin E₂, collagenase, neutral proteinase and reactive oxygen intermediates.

T-cell-mediated hypersensitivity as a pathogenic factor

- Suppression of disease by cyclosporin or anti-CD4 treatment is strong evidence for T-cell involvement. So is an HLA-linked risk factor.
- There is a prevailing view that organ-specific inflammatory lesions are caused by autoreactive pathogenic Th1 cells.
- Activated T-cells are abundant in the rheumatoid synovium and their production of TNF and GM-CSF complements the immune complex stimulus for pannus formation.
- RA patients have poor corticosteroid responses to triggering of the pituitary–adrenal feedback loop and treatment with low, virtually maintenance doses of steroids is beneficial.
- Thyrocytes expressing MHC class II in autoimmune

thyroid disease are direct targets for locally activated Th1 cells specific for thyroid peroxidase.

- That autoimmunity can cause thyroiditis is further shown by the deliberate induction of disease in rodents through immunization with thyroid antigens in complete Freund's adjuvant.
- The onset of IDDM is delayed by cyclosporin, HLA-DQ risk factors are prominent, and T-cell proliferative responses to β -islet cell antigens reflect prognosis of disease.
- Th1 cells from diseased NOD mice, which mimic the human disorder in histopathology and autoimmunity, can produce typical pancreatic lesions in young mice of the same strain. Introduction of a transgene encoding changes at residues 56 or 57 in the H-2 β chain dramatically ameliorates disease.
- Similarity to experimental allergic encephalomyelitis, a demyelinating disease induced by immunization with myelin in complete Freund's adjuvant, has made autoimmunity the front-running hypothesis in MS. Approximately one-third of the IL-2 or -4 activatable T-cells in the CSF of MS patients are specific for myelin and the DR2 phenotype is a strong risk factor.

(continued p. 448)

Some systemic disorders with a vascular component of unknown pathogenesis

- Immunologically mediated vascular lesions are of central importance in Wegener's granulomatosis, temporal arteritis, scleroderma and atherosclerosis.

Diagnostic value of autoantibody tests

- A wide range of serum autoantibodies now provide valuable diagnostic markers.
- Routine immunofluorescent screening is carried out on composite sections of human thyroid and stomach and rat kidney and liver, supplemented by agglutination tests for rheumatoid factors, thyroid and red cell antibodies, and by radioimmunoassays for intrinsic factor and acetylcholine receptor antibodies.
- Solid-phase ELISA tests are used for antibodies to DNA and other nuclear antigens and will increasingly displace fluorescence as purified autoantigens become available.

Treatment of autoimmune disorders

- Therapy conventionally involves metabolic control and

the use of anti-inflammatory and immunosuppressive drugs.

- A whole variety of potential immunological control therapies are under intensive investigation. These include antibody and T-cell idiotype manipulations and attempts to induce antigen-specific unresponsiveness particularly to T-cells using peptides administered parenterally, orally or nasally.
- Organ-related bystander tolerance means that single epitopes can induce suppression of pathogenic cells within an organ reacting to other epitopes on the same or other antigens.
- Plasma exchange may be of value especially in combination with antimetabolic drugs.
- The accompanying comparison of organ-specific and nonorgan-specific autoimmune disorders (table 20.3) gives an overall view of many of the points raised in the last two chapters.

See the accompanying website (www.roitt.com) for multiple choice questions.

Table 20.3. Comparison of organ-specific and nonorgan-specific diseases.

ORGAN-SPECIFIC (e.g. THYROIDITIS, GASTRITIS, ADRENALITIS)	NONORGAN-SPECIFIC (e.g. SYSTEMIC LUPUS ERYTHEMATOSUS)
DIFFERENCES	
Antigens only available to lymphoid system in low concentration	Antigens accessible at higher concentrations
Antibodies and lesions organ-specific	Antibodies and lesions nonorgan-specific
Clinical and serologic overlap – thyroiditis, gastritis and adrenalitis	Overlap SLE, rheumatoid arthritis, and other connective tissue disorders
Familial tendency to organ-specific autoimmunity	Familial connective tissue disease
Lymphoid invasion, parenchymal destruction by Th1 cell-mediated hypersensitivity and/or antibodies	Lesions largely due to deposition of antigen-antibody complexes but may also have Th1 component as in RA
Therapy aimed at controlling metabolic deficit or tolerizing T-cells	Therapy aimed at inhibiting inflammation and antibody synthesis
Tendency to cancer in organ	Tendency to lymphoreticular neoplasia
Antigens evoke organ-specific antibodies in normal animals with complete Freund's adjuvant	No antibodies produced in animals with comparable stimulation
Experimental lesions produced with antigen in Freund's adjuvant but also have spontaneous disease models	Diseases and autoantibodies arise spontaneously in certain animals (e.g. NZB mice and hybrids)
SIMILARITIES	
Circulating autoantibodies react with normal body constituents	
Patients often have increased immunoglobulins in serum	
Antibodies may appear in each of the main immunoglobulin classes particularly IgG and are usually high affinity and mutated	
Greater incidence in women	
Disease process not always progressive; exacerbations and remissions	
Association with HLA	
Spontaneous diseases in animals genetically programmed	
Autoantibody tests of diagnostic value	

FURTHER READING

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CD markers

CD (Cluster of differentiation) markers on hematopoietic cells.

CD	Identity	Function	Cellular distribution	Ligands
CD1a	Non-MHC, class I-like molecule	Presentation of lipid & glycolipid antigens	Thymocyte, DC, LC, B(sub)	Lipid & glycolipid antigens, $\alpha\beta$ TCR
CD1b	Non-MHC, class I-like molecule	Presentation of lipid & glycolipid antigens	Thymocyte, DC, LC, B(sub)	Lipid & glycolipid antigens, $\alpha\beta$ TCR
CD1c	Non-MHC, class I-like molecule	Presentation of lipid & glycolipid antigens	Thymocyte, DC, LC, B(sub)	Lipid & glycolipid antigens, $\alpha\beta$ TCR
CD1d	Non-MHC, class I-like molecule	Presentation of lipid & glycolipid antigens	Leukocytes, epithelium	Lipid & glycolipid antigens, $\alpha\beta$ TCR
CD1e	Non-MHC, class I-like molecule	Presentation of lipid & glycolipid antigens	Unknown	Lipid & glycolipid antigens, $\alpha\beta$ TCR
CD2	LFA-2, T11	T cell adhesion to target cells or APC. T cell activation. Regulation of cytolysis	Thymocyte, T, NK	CD58 (LFA-3), CD59, CD48, CD15
CD3 δ	TCR-associated molecule	T cell activation	Thymocyte, T	—
CD3 ϵ	TCR-associated molecule	T cell activation	Thymocyte, T	—
CD3 γ	TCR-associated molecule	T cell activation	Thymocyte, T	—
CD4	T4, L3T4	Co-receptor for MHC class II-restricted antigen-induced T cell activation	Thymocyte (sub), T(sub), Mo, M ϕ	MHC class II, HIV gp120
CD5	Leu-1, Ly-1	Modulates signaling through TCR & BCR	Thymocyte, T, B(sub)	CD72
CD6	T12	T cell activation & thymocyte-stromal cell interaction	Thymocyte, T, B(sub)	CD166 (ALCAM)
CD7	Tp41	Regulation of cytokine production & sensitivity to LPS-induced shock syndromes	SC, Thymocyte, T, NK	Unknown
CD8 α	T8, Lyt2/3	Co-receptor for MHC class I-restricted antigen-induced T cell activation	Thymocyte (sub), T(sub)	MHC class I
CD8 β	T8, Lyt2/3	Co-receptor for MHC class I-restricted antigen-induced T cell activation	Thymocyte (sub), T(sub)	MHC class I
CD9	DRAP27	Platelet activation & aggregation	Pre-B, Plt, *B, *T, eosinophils, basophils, endothelium, epithelium	Pregnancy-specific glycoprotein-17 (PSG17)
CD10	CALLA, NEP	Neutral endopeptidase	BM	Unknown

(Continued on p. 452)

CD markers (*continued*)

CD	Identity	Function	Cellular distribution	Ligands
CD11a	LFA-1 α chain, α_L integrin chain	Leukocyte–leukocyte/leukocyte–endothelium interactions. Co-stimulation	All leukocytes	CD54 (ICAM-1), CD102 (ICAM-2), CD50 (ICAM-3), CD242 (ICAM-4)
CD11b	Mac-1 α chain, α_M integrin chain, CR3	Adhesion to vascular endothelium. Complement receptor	NK, Gr, Mo, T(sub), B(sub)	CD54 (ICAM-1), CD102 (ICAM-2), CD242 (ICAM-4), iC3b, C3dg, C3d, fibrinogen
CD11c	α_x integrin chain, CR4	Adhesion to vascular endothelium. Complement receptor	NK, Gr, Mo, M ϕ	iC3b, C3dg, fibrinogen
CDw12	Phosphoprotein p90–120	Unknown	Mo, Gr, NK	Unknown
CD13	Aminopeptidase N	Trims peptides bound to MHC class II	Mo, Gr, Endothelium	Unknown
CD14	LPS receptor	Binds LPS, particularly in the presence of LPS-binding protein (LBP). Cellular activation	Mo, M ϕ , Gr	LPS (endotoxin)
CD15	Lewis ^X , 3-FAL, SSEA-1	Adhesion	Neutrophils, Mo, Plt	CD2
CD15s	Sialyl Lewis ^X	Adhesion	Neutrophils, Mo, Plt	CD62E, CD62L, CD62P
CD15u	Sulfated CD15	Adhesion	Neutrophils, Mo, Plt	CD62P
CD16a	Fc γ RIIA	Phagocytosis & ADCC	Gr, NK, M ϕ , DC, B	Fc γ
CD16b	Fc γ RIIIB	Phagocytosis & ADCC	Neutrophils	Fc γ
CDw17	Lactosylceramide	Unknown	Mo, Gr, basophils, Plt, B(sub)	Unknown
CD18	β -chain for CD11, β_2 integrin chain	Adhesion & signaling	All leukocytes	see CD11a,b,c
CD19	B4	Component of CD19/CD21/CD81/CD225 signal transduction complex. B cell activation & proliferation	B, FDC	Heparin/heparan sulphate, IgM
CD20	B1, Bp35	B cell activation & proliferation	B	Unknown
CD21	CR2, EBV-R, C3dR	Complement receptor. Component of CD19/CD21/CD81/CD225 signal transduction complex. B cell activation & proliferation	B, FDC	iC3b, C3d, C3dg, CD23, EBV
CD22	BL-CAM	B cell adhesion & signaling	B(mat)	Sialyl proteins, CD45
CD23	Fc ϵ RII, low affinity IgE receptor	Regulation of IgE production. Adhesion and signaling	B, Mo, FDC	Fc ϵ , CD21
CD24	HSA, BA-1	Apoptotic signaling	B, Gr(mat)	CD62P
CD25	IL-2R α -chain	Cytokine receptor	*T, *B, *Mo, *M ϕ	IL-2
CD26	DPPIV	Endopeptidase. Costimulation for T cell activation	*T, B, M ϕ , NK, thymocytes	Adenosine deaminase
CD27	S152, T14	Costimulation signal for T & B cell activation	Thymocytes, T, B(sub), NK	CD70
CD28	T44	Costimulatory molecule	Thymocytes, T, *B	CD80, CD86
CD29	VLA β -chain	Adhesion	Leukocytes, Plt, erythrocytes	See CD49a–f & CD51. CD171 (in mouse)
CD30	Ber-H2	Costimulatory for T cells	*T, *B, *NK, Mo	CD153

(Continued)

CD markers (*continued*)

CD	Identity	Function	Cellular distribution	Ligands
CD31	PECAM-1	Transendothelial migration of leukocytes	Mo, Plt, neutrophils, NK, endothelium, T(sub)	CD31 (homophilic), CD51/CD61 ($\alpha_v\beta_3$ integrin), CD38
CD32	Fc γ RII	Phagocytosis & ADCC	B, M ϕ , Gr, Mo, FDC, NK	Fc γ
CD33	gp67	Adhesion	Mo, M ϕ , mast cells	Sialic acid on sialo-glycoconjugates
CD34	gp105–120	Adhesion	SC, early lymphohematopoietic progenitor cells, endothelium	CD62L
CD35	CR1, C3b/C4bR	Complement receptor. Accelerates decay of C3 & C5 convertases. Facilitates phagocytosis	Erythrocytes, neutrophils, Mo, B, T(sub), eosinophils, FDC	C3b, C4b, iC3b
CD36	GPIIb, MFGM, PAS IV	Adhesion. Scavenger receptor. Phagocytosis of apoptotic cells	Endothelium, Plt, Mo, M ϕ , erythrocytes	Thrombospondin, collagen, oxidized LDL, long chain fatty acids, anionic phospholipids
CD37	gp52–40	Signal transduction	B, T, neutrophils, Mo	Unknown
CD38	T10	Like CD157, has ADP-ribosyl cyclase/cyclic ADP-ribose hydrolase activity. Modulates cell activation. Adhesion	Most leukocytes	CD31, hyaluronic acid
CD39	gp80	Adhesion	B(sub), *NK(sub), *T(sub)	CD39 (homophilic)
CD40	Bp50	B cell activation, proliferation & differentiation	B, M ϕ , DC, FDC, endothelium	CD154
CD41	α IIb integrin chain	Platelet aggregation	Plt, megakaryocytes	VWF, fibrinogen, fibronectin
CD42a	GPIX	Plt adherence & aggregation at sites of vascular damage	Plt, megakaryocytes	—
CD42b	GP1b α	Plt adherence & aggregation at sites of vascular damage	Plt, megakaryocytes	VWF, thrombin
CD42c	GP1b β	Plt adherence & aggregation at sites of vascular damage	Plt, megakaryocytes	—
CD42d	GPV	Plt adherence & aggregation at sites of vascular damage	Plt, megakaryocytes	—
CD43	Leucosialin, gp115	Anti-adhesion & pro-adhesion. Cellular activation	Most leukocytes	Sialoadhesin (Siglec-1)
CD44	HERMES, Pgp-1, H-CAM	Leukocyte attachment & rolling on endothelium. Homing to peripheral lymphoid organs & sites of inflammation. T cell activation	Widespread	Hyaluronan, osteopontin, ankyrin, fibronectin
CD44R	Variant isoform of CD44	Leukocyte attachment & rolling on endothelium. Homing to peripheral lymphoid organs & sites of inflammation	Mo, *T, *B, epithelium	Hyaluronan, osteopontin, ankyrin, fibronectin
CD45	Leukocyte common antigen (LCA), B220	Lymphocyte signaling	All leukocytes	Galectin-1, CD22
CD45RA	Restricted LCA	Lymphocyte signaling	Thymocytes, T(sub), B, Gr(sub), Mo, NK	Galectin-1, CD22

(Continued on p. 454)

CD markers (continued)

CD	Identity	Function	Cellular distribution	Ligands
CD45RB	Restricted LCA	Lymphocyte signaling	Thymocytes, T(sub), B, Gr, Mo, NK	Galectin-1, CD22
CD45RC	Restricted LCA	Lymphocyte signaling	Thymocytes, NK, T(sub), B	Galectin-1, CD22
CD45RO	Restricted LCA	Lymphocyte signaling	Thymocytes, T(sub), Gr, Mo, DC	Galectin-1, CD22
CD46	Membrane cofactor protein (MCP)	Limits formation & function of C3 convertases	Widespread	Serum factor I protease, C3b, C4b, measles virus
CD47	Integrin-associated protein (IAP)	Adhesion. Thrombospondin receptor	Widespread	Thrombospondin, CD172a
CD47R	MEM133, CDw149	Adhesion. Thrombospondin receptor	T, B, Mo	Thrombospondin
CD48	Blast-1, OX-45, BCM1	T cell adhesion to target cells & APC	Leukocytes (except neutrophils)	CD2
CD49a	VLA-1 α chain, α_1 integrin chain	Receptor for laminin & collagen	Fibroblasts, capillary endothelium, NK, *T	Laminin, collagen I & IV
CD49b	VLA-2 α chain, α_2 integrin chain	Regulates expression of MMP-1 & collagen type I	Fibroblasts, endothelium, Plt, B, T, keratinocytes	Collagen I-IV, laminin
CD49c	VLA-3 α chain, α_3 integrin chain	Adhesion	Fibroblasts, keratinocytes, epithelium, B	Fibronectin, collagen
CD49d	VLA-4 α chain, α_4 integrin chain	Leukocyte rolling, adhesion & migration	Widespread	CD106 (VCAM-1), fibronectin
CD49e	VLA-5 α chain, α_5 integrin chain	Cell adhesion, migration & matrix assembly	Fibroblasts, epithelium, endothelium, muscle, Plt	Fibronectin, CD171 (in mouse)
CD49f	VLA-6 α chain, α_6 integrin chain	Cell adhesion, spreading & migration	Widespread	Laminin
CD50	ICAM-3	Costimulatory molecule	Leukocytes, LC, endothelium	$\alpha_d\beta_2$ integrin, CD11a/CD18 (LFA-1), CD209
CD51	α_v chain of vitronectin receptor	Recruitment, distribution & retention of cells via extracellular matrix	Endothelium, Mo, Plt, osteoclasts, B(sub), T	CD31, CD171, laminin, fibrinogen, fibronectin
CD52	Campath-1	Unknown	Thymocytes, T, B, Mo, M ϕ	Unknown
CD53	MEM-53	Activation	Leukocytes	Unknown
CD54	ICAM-1	Leukocyte adhesion to endothelium in inflammation	Endothelium, *T, *B, epithelium, Mo	CD11a/CD18 (LFA-1), CD11b/CD18 (Mac-1), CD227, rhinovirus, <i>Plasmodium falciparum</i> -infected erythrocytes
CD55	Decay accelerating factor (DAF)	Limits formation & half life of C3 convertases	Widespread	C3 convertases, CD97
CD56	NCAM, D2-CAM, Leu-19, NKH1	Adhesion	NK, T(sub), brain	CD56 (homophilic), heparan sulfate
CD57	HNK-1, Leu-7	Non-MHC restricted cytotoxicity after activation	NK, T(sub), B(sub), Mo	Unknown
CD58	LFA-3	T cell adhesion to target cells & APC	Widespread	CD2

(Continued)

CD markers (continued)

CD	Identity	Function	Cellular distribution	Ligands
CD59	HRF20, MIRL	Inhibits formation of membrane attack complex. T cell adhesion to target cells & APC	Widespread	C8, C9, CD2
CD60a	GD3	Costimulation of T cells	T(sub), Plt	Sialic acid on sialo-glycoconjugates
CD60b	9-O-acetyl-GD3	Costimulation of T cells	T(sub), Plt	Sialic acid on sialo-glycoconjugates
CD60c	7-O-acetyl-GD3	Costimulation of T cells	T(sub), Plt	Sialic acid on sialo-glycoconjugates
CD61	Vitronectin receptor β chain, β_3 integrin chain	Adhesion	Plt, megakaryocytes, Mo, M ϕ , endothelium	CD31, CD171
CD62E	E-selectin, ELAM-1	Tethering & rolling of leukocytes on cytokine-activated endothelium	Endothelium	ESL-1, CD15s (sialyl Lewis ^X), sialyl Lewis ^a , CLA, CD66a, CD66c, CD162, CD227
CD62L	L-selectin, Leu-8	Tethering & rolling of lymphocytes on LN HEV & leukocytes on endothelium	T, B, Mo, Gr, NK. Downregulated on activation by endoproteolysis (shedding)	CD34, MAdCAM, GlyCAM-1, CD15s (sialyl Lewis ^X), sialyl Lewis ^a
CD62P	P-selectin, GMP-140, PADGEM	Adhesion of Plt & leukocytes to endothelium	Thrombin/histamine activated Plt & endothelium. Stored in granules prior to activation	CD162, CD24, CD15s (sialyl Lewis ^X), sialyl Lewis ^a
CD63	LIMP, ME491	Unknown	*Plt, *Endothelium	Unknown
CD64	Fc γ RI	Phagocytosis & ADCC	Mo, M ϕ , DC	Fc γ
CD65	VIM-8, VIM-11, asialylated ceramide dodecasaccharide 4c	Unknown	Gr	Unknown
CD65s	VIM-2, sialylated ceramide dodecasaccharide 4c	Unknown	Mo, Gr	Unknown
CD66a	NCA-160, BGP (biliary glycoprotein)	Adhesion, neutrophil activation	Gr, Epithelium	CD62E, CD66a (homophilic), c, e
CD66b	NCA-95, CD67	Adhesion, neutrophil activation	Gr	CD66c
CD66c	NCA-50/90	Adhesion, neutrophil activation	Gr, Epithelium	CD62E, CD66a, b, c (homophilic), e
CD66d	CGM1	Neutrophil activation	Gr	Unknown
CD66e	Carcinoembryonic antigen (CEA)	Adhesion	Epithelium	CD66a, c, e (homophilic)
CD66f	SP1, Pregnancy-specific glycoprotein (PSG)	Maintenance of pregnancy?	Placental syncytiotrophoblasts	Unknown
CD68	Microsialin	Unknown	Mo, M ϕ , DC, neutrophils, basophils, mast cells, *T, B(sub)	Oxidized LDL
CD69	AIM, Leu-23	Activation, signal transduction	*T, *B, *NK, *neutrophils, *eosinophils	Unknown
CD70	Ki-24	Costimulation	*T(sub), *B(sub)	CD27
CD71	Transferrin receptor	Iron uptake	*T, *B, *NK, *Mo, M ϕ , SC	Transferrin
CD72	Lyb-2	B cell activation & proliferation	B	CD5, CD100
CD73	Ecto-5'-nucleotidase	Unknown	T(sub), B(sub), FDC, endothelium, epithelium	Adenosine monophosphate

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CD markers (continued)

CD	Identity	Function	Cellular distribution	Ligands
CD74	Invariant chain	Intracellular sorting of MHC class II molecules	B, M ϕ , *T, *endothelium, *epithelium	MHC class II
CD75	Non-sialylated/masked lactosamine epitopes	Unknown	B, T(sub), erythrocytes	Unknown
CD75s	α -2,6-sialylated lactosamines, CDw76	Unknown	B(sub), T(sub), endothelium, epithelium, erythrocytes	Unknown
CD77	BLA, globotriaosylceramide	Unknown	Germinal center B	Shiga toxin, verotoxin-1
CD79a	Ig α , mb1	Signal transduction as part of B cell receptor	B	—
CD79b	Ig β , B29	Signal transduction as part of B cell receptor	B	—
CD80	B7.1, BB1	Regulates T cell activation	*B, *T, M ϕ , DC	CD28, CD152
CD81	TAPA-1	Component of CD19/CD21/CD81/CD225 signal transduction complex	Leukocytes (except neutrophils), endothelium, epithelium	—
CD82	R2, IA4	Signal transduction	Leukocytes	Unknown
CD83	HB15	Unknown	DC, LC	Unknown
CD84	2G7	Costimulation	Mo, M ϕ , Plt, B, T(sub), thymocytes	CD84 (homophilic)
CD85	ILT/LIR family	Inhibition/activation of cytotoxicity	B, T(sub), NK(sub), Mo, M ϕ , DC, Gr	HLA-G, some HLA-A & -B alleles
CD86	B7.2	Regulates T cell activation	B, Mo, DC, LC	CD28, CD152
CD87	Urokinase plasminogen activator (uPA) receptor	Inflammatory cell invasion	Mo, M ϕ , *T, endothelium, NK, neutrophils	Vitronectin, uPA
CD88	C5a receptor	Complement receptor	Gr, Mo, DC	C5a
CD89	Fc α R	Induces phagocytosis, degranulation & respiratory burst	Gr, Mo, M ϕ , *eosinophils	IgA1, IgA2
CD90	Thy-1	T cell activation	Thymocytes, T (mouse), SC, HEV	Unknown
CD91	α_2 -macroglobulin receptor, LDLR-related protein	Receptor	Mo, M ϕ	α_2 -macroglobulin
CD92	VIM15	Unknown	Gr, Mo	Unknown
CD93	p120	C1q receptor	Mo, Gr, endothelium	C1q
CD94	kp43	Inhibition/activation of cytotoxicity	T(sub), NK	Some HLA class I molecules
CD95	Fas, APO-1	Transduces an apoptotic signal	Widespread	CD178 (Fas-L)
CD96	Tactile	Adhesion?	*T, *NK	Unknown
CD97	BL-KDD/F12	Unknown	Gr, Mo, M ϕ , DC, *T, *B	CD55
CD98	4F2	Cellular activation	Widespread	Unknown
CD99	E2, MIC2	Apoptotic signaling	Thymocytes, T, B, erythrocytes	Unknown
CD100	SEMA4D	Lymphocyte proliferation	All leukocytes	CD72
CD101	p126	Costimulation	Gr, Mo, M ϕ , DC, *T	Unknown
CD102	ICAM-2	Lymphocyte recirculation & trafficking, Costimulation	Endothelium, neutrophils, lymphocytes, Mo, Plt	CD11a/CD18 (LFA-1)

(Continued)

CD markers (*continued*)

CD	Identity	Function	Cellular distribution	Ligands
CD103	α_E -integrin chain, HML-1	Adhesion of mucosal lymphocytes to epithelium	Most intraepithelial T & 50% of lamina propria lymphocytes. Absent from PBLs	E-cadherin
CD104	β_4 -integrin chain	Adhesion molecule with α_6 integrin chain	Epithelium	Laminin, epiligrin
CD105	Endoglin	Regulatory component of TGF β receptor	*Mo, M ϕ , endothelium	TGF β
CD106	VCAM-1	Leukocyte migration & recruitment to sites of inflammation	M ϕ , DC, FDC, BM stromal cells, endothelium	CD49d/CD29 (VLA4), $\alpha_4\beta_7$ integrin
CD107a	LAMP-1	Unknown	*Plt, *T, *endothelium, *neutrophils	Unknown
CD107b	LAMP-2	Unknown	*Plt, *endothelium, *neutrophils	Unknown
CD108	JMH blood group antigen	Adhesion	T, B, erythrocytes	Unknown
CD109	Sialomucin	Unknown	Endothelium, stromal cells, Plt, *T	Unknown
CD110	MPL	Thrombopoietin receptor	Plt, megakaryocytes	Thrombopoietin
CD111	PRR1/Nectin1	Adhesion	Widespread	Herpes simplex virus, CD111 (homophilic)
CD112	PRR2/Nectin2	Adhesion	Widespread	Herpes simplex virus, CD112 (homophilic)
CD114	G-CSF receptor	Growth factor receptor	Gr, Mo, Plt, endothelium	G-CSF
CD115	CSF-1 receptor	Growth factor receptor	Mo, M ϕ	M-CSF
CD116	GM-CSF receptor α -chain	Growth factor receptor	Mo, M ϕ , neutrophils, eosinophils, DC	GM-CSF
CD117	Stem cell factor (SCF) receptor, <i>c-kit</i>	Growth factor receptor	SC, hematopoietic progenitors, mast cells	SCF
CDw119	IFN γ receptor	Cytokine receptor	Mo, Gr	IFN γ
CD120a	TNF receptor type I, p55	Cytokine receptor	Widespread	TNF (TNF α), lymphotoxin (TNF β)
CD120b	TNF receptor type II, p75	Cytokine receptor	Widespread	TNF (TNF α), lymphotoxin (TNF β)
CD121a	IL-1 RI	Cytokine receptor	Thymocytes, T, fibroblasts, endothelium	IL-1 α , β , ra
CD121b	IL-1 RII	Cytokine receptor	B, T, Mo, M ϕ	IL-1 α , β , ra
CD122	IL-2 & IL-15 receptor β -chain	Cytokine receptor	T, B, Mo, M ϕ , NK	IL-2, IL-15
CD123	IL-3 receptor α -chain	Cytokine receptor	Hematopoietic precursors	IL-3
CD124	IL-4 & IL-13 receptor α -chain	Cytokine receptor	T, B, hematopoietic precursors, fibroblasts	IL-4, IL-13
CDw125	IL-5 receptor α -chain	Cytokine receptor	Eosinophils, basophils, *B	IL-5
CD126	IL-6 receptor α -chain	Cytokine receptor	*B, plasma cells, T, Mo, epithelium, fibroblasts, neural cells	IL-6

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CD markers (continued)

CD	Identity	Function	Cellular distribution	Ligands
CD127	IL-7 receptor α -chain	Cytokine receptor	Immature thymocytes, hepatocytes, pre-B, T(mat)	IL-7
CDw128a	IL-8 receptor α -chain, CXCR1	Chemokine receptor	Gr, T(sub), Mo, endothelium	CXCL8 (IL-8), CXCL6 (GCP-2)
CDw128b	IL-8 receptor β -chain, CXCR2	Chemokine receptor	Gr, T(sub), Mo, endothelium	CXCL8 (IL-8), CXCL1 (GRO α), CXCL2 (GRO β), CXCL3 (GRO γ), CXCL5 (ENA-78), CXCL6 (GCP-2), CXCL7 (NAP-2)
CD130	gp130 component of IL-6, IL-11, LIF, OSM, CNTF & CT-1 receptors	Signal transduction	Widespread	Oncostatin M (OSM) is the only direct ligand
CDw131	Common β -chain of IL-3, IL-5 & GM-CSF receptors	Signal transduction	Mo, Gr, B	Not involved in cytokine binding
CD132	Common γ -chain of IL-2, 4, 7, 9 & 15 receptors	Cytokine binding & signaling	T, B, NK, Mo, M ϕ , neutrophils, fibroblasts	IL-2, 4, 7, 9 & 15
CD133	AC133	Unknown	SC	Unknown
CD134	OX40	Adhesion of *T to vascular endothelium	*T	OX40 ligand, gp34
CD135	Flt3/Flk2	Receptor, tyrosine kinase	Myelomonocytic & B progenitors	Flt3/Flk2 ligand
CD136	MSP receptor, RON	Receptor, tyrosine kinase	Epithelium, some hematopoietic cells	Macrophage stimulating protein, HGF1
CDw137	4-1BB	Costimulatory molecule for T cell proliferation	T	4-1BB ligand
CD138	Syndecan-1	Heparan sulfate proteoglycan	B(sub)	Collagen type 1
CD139	—	Unknown	B, Mo, Gr	Unknown
CD140a	PDGF receptor α	Tyrosine kinase	Endothelium, stromal cells, mesangial cells	PDGF A
CD140b	PDGF receptor β	Tyrosine kinase	Endothelium, stromal cells, mesangial cells	PDGF B
CD141	Thrombomodulin	Downregulates coagulation	Megakaryocytes, Plt, Mo, neutrophils, endothelium, smooth muscle	Thrombin
CD142	Tissue factor, thromboplastin	Induces coagulation	Mo, endothelium, epithelium, keratinocytes	Factor VIIa
CD143	ACE, peptidyl-dipeptidase A	Metabolism of angiotensin & bradykinin	Endothelium, epithelium, M ϕ	Angiotensin, bradykinin
CD144	VE-cadherin	Adhesion	Endothelium	Unknown
CDw145	—	Unknown	Endothelium	Unknown
CD146	MUC18, S-endo	Adhesion?	Endothelium, T	Unknown
CD147	Neurothelin, basigen, EMMPRIN, OX-47	Adhesion?	Leukocytes, endothelium, erythrocytes	Unknown
CD148	HPTP-eta, DEP-1	Tyrosine phosphatase RPTPase type III	Widespread	Tyrosine phosphorylated proteins

(Continued)

CD markers (*continued*)

CD	Identity	Function	Cellular distribution	Ligands
CD150	SLAM	Signaling	T(sub), B, thymocytes, DC, endothelium	CD150 (homophilic), measles virus
CD151	PETA3	Unknown	Megakaryocytes, Plt, Gr, endothelium, smooth muscle, epithelium	Unknown
CD152	CTLA-4	Negative regulator of T cell costimulation	*T	CD80, CD86
CD153	CD30L	Costimulatory for T cells	*T, Gr, B, Mφ	CD30
CD154	CD40L, TRAP	Costimulatory	*T, *B, NK, mast cells	CD40
CD155	Poliovirus receptor	Adhesion?	Mo, Mφ, thymocytes, CNS neurons	Vitronectin, poliovirus
CD156a	ADAM8	Extravasation of leukocytes?	Neutrophils, Mo	Unknown
CD156b	TACE (TNFα converting enzyme), ADAM17	Protease which releases soluble forms of TNF from cells	Widespread	Unknown
CD157	Bst-1	Like CD38, has ADP-ribosyl cyclase/cyclic ADP-ribose hydrolase activity	Mo, Mφ, neutrophils, endothelium, T(sub), B(sub)	NAD, cyclic ADP-ribose
CD158	KIR family	Inhibition/activation of cytotoxicity	NK, T(sub)	Various MHC class I molecules
CD159a	NKG2A	Inhibition of cytotoxicity	NK, T(sub)	HLA class I signal sequence derived peptides presented by HLA-E
CD160	BY55, NK1	Costimulation	NK, CD8 ⁺ T	HLA-G
CD161	NKRP1A	Activation?	NK, T(sub)	Ganglioside GM2
CD162	BY55, PSGL-1 (P-selectin glycoprotein ligand-1)	Adhesion	Mo, Gr, T, B(sub)	CD62E, CD62P
CD162R	PEN5, PSGL-1 (P-selectin glycoprotein ligand-1)	Adhesion	NK, *T	CD62P
CD163	M130	Hemoglobin scavenger receptor	Mo, some Mφ	Haptoglobin-hemoglobin complexes
CD164	MGC-24	Adhesion of hematopoietic progenitor cells to stromal cells	Myeloid cells, T, epithelium, BM stromal cells, hematopoietic progenitor cells	Unknown
CD165	AD2, gp37	Adhesion of thymocytes to thymic epithelium	Plt, thymocytes, lymphocytes (sub), thymic epithelium, Mo, CNS neurons	Unknown
CD166	ALCAM, BEN	Adhesion	T, *Mo, epithelium, fibroblasts, neurons	CD6
CD167a	DDR1 (Discoid domain receptor 1)	Receptor tyrosine kinase	Epithelial cells	Collagen
CD168	RHAMM	Cell migration	Fibroblasts	Hyaluronan
CD169	Sialoadhesin, SIGLEC-1	Cell adhesion	Mφ	Sialic acid on sialo-glycoconjugates
CD170	SIGLEC-5	Cell adhesion	Neutrophils	Sialic acid
CD171	L1 (NCAM-L1)	Cell adhesion, costimulation	CD4 ⁺ T, B(sub), Mo, DC, FDC, fibroblasts, neurons	Laminin, Neurocan, Phosphocan, CD51/CD61 (α _v β ₃ integrin) (and CD49e/CD29 α ₅ β ₁ integrin in mouse)

(Continued on p. 460)

CD markers (*continued*)

CD	Identity	Function	Cellular distribution	Ligands
CD172a	PTPNS1, SHPS1, SIRP α	Adhesion	Mo, M ϕ , DC	CD47
CD173	Blood group H type 2	Glycocalyx component	Widespread	Unknown
CD174	Lewis ^x	Adhesion	Widespread	Unknown
CD175	Tn	Mucin-type carbohydrate	Epithelium. Elevated expression in some tumors	Unknown
CD175s	Sialyl-Tn	Mucin-type carbohydrate	Epithelium. Elevated expression in some tumors	Unknown
CD176	TF (Thomsen–Friedenreich antigen)	Mucin-type carbohydrate	Epithelium, M ϕ (sub). Elevated expression in some tumors	Unknown
CD177	NB1, HNA-2a	Receptor?	Gr	Unknown
CD178	Fas ligand	Induction of apoptosis	*T	CD95 (Fas)
CD179a	Vpre-B, surrogate light chain	B cell differentiation	Pro-B, early pre-B	Unknown
CD179b	λ 5, surrogate light chain	B cell differentiation	Pro-B, early pre-B	Unknown
CD180	RP105, Bgp95	Controls B cell recognition of, & signaling by, LPS	B, Mo, DC	Unknown
CD183	CXCR3	Chemokine receptor, signaling	Effector / memory T, B(sub), NK(sub), DC(sub), eosinophils	CXCL9 (Mig), CXCL10 (IP-10), CXCL11 (I-TAC)
CD184	CXCR4	Chemokine receptor, signaling	T, B, DC, Mo	CXCL12 (SDF-1 α/β)
CD195	CCR5	Chemokine receptor, signaling	T, NK, DC, Mo	CCL3 (MIP-1 α), CCL4 (MIP-1 β), CCL5 (RANTES)
CDw197	CCR7	Chemokine receptor, signaling	B, T, DC	CCL19 (MIP-3 β), CCL21 (6Ckine)
CD200	OX2	Downregulates M ϕ activation	Widespread	OX2R
CD201	EPCR	Protein C receptor	Endothelial cells	Protein C
CD202b	TEK, Tie2	Receptor tyrosine kinase	Endothelial cells	Angiopoietin-1
CD203c	E-NPP3, PDNP3	Enzyme involved in hydrolysis of extracellular nucleotides	Basophils, mast cells	Nucleoside di- and tri-phosphates
CD204	Macrophage scavenger receptor-1	Antigen uptake	M ϕ	Negatively charged macromolecules
CD205	DEC205	Antigen uptake	DC, B, thymic epithelium	Unknown
CD206	Macrophage mannose receptor	Antigen uptake	M ϕ , liver endothelial cells	Oligomannose-containing carbohydrates
CD207	Langerin	Antigen uptake	LC, DC	Mannose
CD208	DC-LAMP (Dendritic cell lysosome-associated membrane protein)	Regulation of MHC class II antigen processing compartments	DC	Unknown
CD209	DC-SIGN (DC-specific ICAM-3- grabbing nonintegrin)	Adhesion	DC	CD50 (ICAM-3)
CDw210	IL-10 receptor	Cytokine receptor	T, B, NK, Mo, M ϕ	IL-10
CD212	IL-12 receptor	Cytokine receptor	*T, *NK	IL-12
CD213a1	IL-13 receptor α 1	Cytokine receptor	B, Mo, fibroblasts, endothelial cells	IL-13

(Continued)

CD markers (continued)

CD	Identity	Function	Cellular distribution	Ligands
CD213a2	IL-13 receptor $\alpha 2$	Cytokine receptor	B, Mo, fibroblasts, endothelial cells	IL-13
CDw217	IL-17 receptor	Cytokine receptor	Widespread	IL-17
CD220	Insulin receptor	Hormone receptor	Muscle, adipose tissue, hepatocytes	Insulin
CD221	Insulin-like growth factor 1 (IGF-1) receptor	Cell survival	Bone, cartilage, muscle, adipose tissue	IGF-1
CD222	Mannose 6-phosphate receptor, insulin-like growth factor II receptor	Lysosomal sorting & internalization	Widespread	Molecules bearing mannose 6-phosphate
CD223	Lymphocyte activation gene 3 (LAG-3)	Lymphocyte activation	*T, *NK	MHC class II
CD224	γ -glutamyl transpeptidase (GGT)	Involved in uptake of extracellular glutathione	T, B, liver, kidney	Glutathione
CD225	Leu13	Component of CD19/CD21/CD81/CD225 signal transduction complex	Widespread	—
CD226	DNAM-1, PTA1	Adhesion, activation	NK, Plt, Mo, T(sub), B(sub)	Unknown
CD227	MUC1	Modulation of adhesion, signaling	T, DC, Mo, B, FDC, glandular and ductal epithelium	CD54, CD62E, Grb2
CD228	Melanotransferrin	Iron transport	Melanoma cells	Iron
CD229	Ly9	Adhesion	T, B	Unknown
CD230	PrP (prion protein)	Signal transduction?	Widespread	Unknown (plasminogen binds to the misfolded form of PrP)
CD231	TALLA-1, A15	Unknown	T cell ALL	Unknown
CD232	Plexin C1	Receptor for VESP	B	Virally-encoded semaphorin (VESP)
CD233	Band 3	Anion exchange	Erythrocytes	Chloride and bicarbonate
CD234	Duffy antigen receptor for chemokines (DARC)	Chemokine receptor	Erythrocytes, endothelium, epithelium, neurons	CXCL8 (IL-8), CCL5 (RANTES), CCL2 (MCP-1), <i>Plasmodium vivax</i>
CD235a	Glycophorin A	Mechanical stability & elastic properties of erythrocytes	Erythrocytes, endothelium, epithelium	<i>Plasmodium falciparum</i>
CD235b	Glycophorin B	Mechanical stability & elastic properties of erythrocytes	Erythrocytes, endothelium, epithelium	<i>Plasmodium falciparum</i>
CD235ab	Glycophorin A/B	Mechanical stability & elastic properties of erythrocytes	Erythrocytes, endothelium, epithelium	<i>Plasmodium falciparum</i>
CD236	Glycophorin C/D	Mechanical stability & elastic properties of erythrocytes	Erythrocytes, endothelium	—
CD236R	Glycophorin C	Mechanical stability & elastic properties of erythrocytes	Erythrocytes, endothelium	—
CD238	Kell blood group	Endopeptidase cleaves the endothelin-3 precursor (big ET3) to produce the multifunctional bioactive peptide ET3	Erythrocytes, myeloid progenitors	Big endothelin-3

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CD markers (*continued*)

CD	Identity	Function	Cellular distribution	Ligands
CD239	B-CAM (B-cell adhesion molecule), Lutheran blood group	Adhesion, signaling?	Widespread	Laminin
CD240CE	Rh blood group CcEe alleles	Transporter?	Erythrocytes	Unknown
CD240D	Rh blood group D antigen	Transporter?	Erythrocytes	Unknown
CD240DCE	Rh blood group DCcEe alleles	Transporter?	Erythrocytes	Unknown
CD241	RHAG (Rh blood group associated glycoprotein)	Ammonium transport	Erythrocytes	Ammonium
CD242	ICAM-4	Adhesion	Erythrocytes	CD11a/CD18 (LFA-1), CD11b/CD18 (Mac-1)
CD243	MDR1	P-glycoprotein drug efflux pump	SC	—
CD244	2B4	Involved in non-MHC restricted cytotoxicity	NK, T(sub)	CD48
CD245	p220/240	Unknown	T	Unknown
CD246	ALK (Anaplastic lymphoma protein kinase)	Signaling	T	Unknown
CD247	ζ chain	TCR & FcγRIIIA signaling	T, Mφ, NK	None

Abbreviations: APC, antigen-presenting cell; BM, bone marrow; T, T-cell; B, B-cell; NK, natural killer cell; Mo, monocyte; Mφ, macrophage; Gr, granulocytes (neutrophils, basophils, eosinophils); Plt, platelet; LC, Langerhans' cell; DC, interdigitating dendritic cell; FDC, follicular dendritic cell; SC, stem cell; *, activated; (mat), mature; (sub), subset; EBV, Epstein-Barr virus; ECM, extracellular matrix; VWF, von Willebrand factor. Further information on each CD antigen can be freely accessed through Protein Reviews on the Web (PROW: Shaw S., Turni L.A. and Katz K.S. [editors], <http://www.ncbi.nlm.nih.gov/prow>). Another excellent reference source is *The Leucocyte Antigen Facts* book (Barclay A.N., Brown M.H., Law S.K.A., McKnight A.J., Tomlinson M.G. and van der Merwe P.A. 2nd edn, 1997, Academic Press, London).

Glossary

acquired immune response: Immunity mediated by lymphocytes and characterized by antigen-specificity and memory.

acute phase proteins: Serum proteins, mostly produced in the liver, which rapidly change in concentration (some increase, some decrease) during the initiation of an inflammatory response.

adjuvant: Any substance which nonspecifically enhances the immune response to antigen.

affinity (intrinsic affinity): The strength of binding (affinity constant) between a receptor (e.g. one antigen-binding site on an antibody) and a ligand (e.g. epitope on an antigen).

allele: Variants of a polymorphic gene at a given genetic locus.

allelic exclusion: The phenomenon whereby, following successful rearrangement of one allele of an antigen receptor gene, rearrangement of the other parental allele is suppressed, thereby ensuring each lymphocyte expresses only a single specificity of antigen receptor (although this does not occur for α chains in T-cells).

allergen: An antigen which causes allergy.

allergy: IgE-mediated hypersensitivity, e.g. asthma, eczema, hayfever and food allergy.

alogeneic: Refers to the genetic differences between individuals of the same species.

allograft: Tissue or organ graft between allogeneic individuals.

allotype: An allelic variant of an antigen which, because it is not present in all individuals, may be immunogenic in members of the same species which have a different version of the allele.

alternative pathway (of complement activation): Activation pathway involving complement components C3, Factor B, Factor D, and Properdin which, in the presence of a stabilizing activator surface such as microbial polysaccharide, generates the alternative pathway C3 convertase $C3bBb$.

anaphylatoxin: A substance (e.g. C3a, C4a or C5a) capable of directly triggering mast cell degranulation.

anaphylaxis: An often fatal hypersensitivity reaction,

triggered by IgE or anaphylatoxin-mediated mast cell degranulation, leading to anaphylactic shock due to vasodilatation and smooth muscle contraction.

anergy: Potentially reversible specific immunological tolerance in which the lymphocyte becomes functionally nonresponsive.

antibody-dependent cellular cytotoxicity (ADCC): A cytotoxic reaction in which an antibody-coated target cell is directly killed by an Fc receptor-bearing leukocyte, e.g. NK cell, macrophage or neutrophil.

antigen: Any molecule capable of being recognized by an antibody or T-cell receptor.

antigen-presenting cell (APC): A term most commonly used when referring to cells that present processed antigenic peptide and MHC class II molecules to the T-cell receptor on CD4⁺ T-cells, e.g. macrophages, dendritic cells, B-cells. Note, however, that most types of cell are able to present antigenic peptides with MHC class I to CD8⁺ T-cells, e.g. as occurs with virally infected cells.

antigenic determinant: A cluster of epitopes (*see* epitope).

apoptosis: A form of programmed cell death, characterized by endonuclease digestion of DNA.

atopic allergy: IgE-mediated hypersensitivity, i.e. asthma, eczema, hayfever and food allergy.

autologous: From the same individual.

avidity (functional affinity): The binding strength between two molecules (e.g. antibody and antigen) taking into account the valency of the interaction. Thus the avidity will always be equal to or greater than the intrinsic affinity (*see* affinity).

β_2 -microglobulin: A 12 kDa protein, not itself encoded within the MHC, but forming part of the structure of MHC class I-encoded molecules.

B-1/B-2 cells: The two major subpopulations of B lymphocytes. B-1 cells bear high levels of surface IgM, lower levels of surface IgD, are CD43⁺, CD23⁻ and most express the cell surface antigen CD5; they are self-renewing, and frequently secrete high levels of antibody which binds to a range of antigens ('poly-specificity') with a relatively low affinity. The major-

- ity of B cells, however, are B-2 which express low levels of surface IgM, higher levels of surface IgD, do not express CD5, and are CD43⁺, CD23⁺; they are directly generated from precursors in the bone marrow, and secrete highly specific antibody.
- basophil:** A type of granulocyte found in the blood and resembling the tissue mast cell.
- BCG (bacille Calmette–Guérin):** Attenuated *Mycobacterium tuberculosis* used both as a specific vaccine for tuberculosis and as an adjuvant.
- biolistics:** The use of small particles, e.g. colloidal gold, as a vehicle for carrying agents (drugs, nucleic acid, etc.) into a cell. Following coating with the desired agent(s), the particles are fired into the dermis of the recipient using a helium-powered gun.
- bispecific antibody:** An artificially produced hybrid antibody in which each of the two antigen-binding arms is specific for a different antigenic epitope. Such antibodies, which can be produced either by chemical cross-linkage or by recombinant DNA techniques, can be used to link together two different antigens or cells, e.g. a cytotoxic T-cell and a tumor cell.
- bursa of Fabricius:** A primary lymphoid organ in avian species, located at the cloacal-hind gut junction; it is the site of B-cell maturation.
- capping:** An active process whereby cross-linking of cell surface molecules (e.g. by antibody) leads to aggregation and subsequent migration of the molecules to one pole of the cell.
- carrier:** Any molecule which when conjugated to a non-immunogenic molecule (e.g. a hapten) makes the latter immunogenic by providing epitopes for helper T-cells which the hapten lacks.
- CD antigen:** Cluster of differentiation designation assigned to leukocyte cell surface molecules which are identified by a given group of monoclonal antibodies.
- CD3:** A trimeric complex of γ , δ and ϵ chains which together with a $\zeta\zeta$ homodimer or $\zeta\eta$ heterodimer acts as a signal transducing unit for the T-cell receptor.
- CD4:** Cell surface glycoprotein, usually on helper T-cells, that recognizes MHC class II molecules on antigen-presenting cells.
- CD8:** Cell surface glycoprotein, usually on cytotoxic T-cells, that recognizes MHC class I molecules on target cells.
- cell-mediated immunity (CMI):** Refers to T-cell mediated immune responses.
- chemokines:** A family of structurally-related cytokines which selectively induce chemotaxis and activation of leukocytes. They also play important roles in lymphoid organ development, cell compartmentalization within lymphoid tissues, Th1/Th2 development, angiogenesis and wound healing.
- chemotaxis:** Movement of cells up a concentration gradient of chemotactic factors.
- chimeric:** Composite of genetically distinct individuals, e.g. following an allogeneic bone marrow graft.
- class switching:** The process by which a B-cell changes the class but not specificity of a given antibody it produces, e.g. switching from an IgM to an IgG antibody.
- classical pathway (of complement activation):** Activation pathway involving complement components C1, C2 and C4 which, following fixation of C1q, e.g. by antigen–antibody complexes, produces the classical pathway C3 convertase C4b2a.
- clonal deletion:** A process by which contact with antigen (e.g. self antigen) at an early stage of lymphocyte differentiation leads to cell death by apoptosis.
- clonal selection:** The selection and activation by antigen of a lymphocyte bearing a complementary receptor, which then proliferates to form an expanded clone.
- clone:** Identical cells derived from a single progenitor.
- colony stimulating factors (CSF):** Factors that permit the proliferation and differentiation of hematopoietic cells.
- complement:** A group of serum proteins, some of which act in an enzymatic cascade, producing effector molecules involved in inflammation (C3a, C5a), phagocytosis (C3b), and cell lysis (C5b-9).
- complementarity determining regions (CDR):** The hypervariable amino acid sequences within antibody and T-cell receptor variable regions which interact with complementary amino acids on the antigen or peptide–MHC complex.
- ConA (concanavalin A):** A T-cell mitogen.
- congenic:** Animals which only differ at a single genetic locus.
- conjugate:** Covalently-linked complex of two or more molecules (e.g. fluorescein conjugated to antibody).
- Coombs' test:** Diagnostic test using anti-immunoglobulin to agglutinate antibody-coated erythrocytes.
- cortex:** Outer (peripheral) layer of an organ.
- C-reactive protein:** An acute phase protein which is able to bind to the surface of microorganisms where it functions as a stimulator of the classical pathway of complement activation, and as an opsonin for phagocytosis.
- cyclophosphamide:** Cytotoxic drug used as an immunosuppressive.
- cyclosporin A:** A T-cell specific immunosuppressive drug used to prevent graft rejection.
- cytokines:** Low molecular weight proteins that stimu-

- late or inhibit the differentiation, proliferation or function of immune cells.
- cytophilic:** Binds to cells.
- cytotoxic:** Kills cells.
- cytotoxic T lymphocyte (Tc):** T-cells (usually CD8⁺) which kill target cells following recognition of foreign peptide–MHC molecules on the target cell membrane.
- delayed-type hypersensitivity (DTH):** A hypersensitivity reaction occurring within 48–72 hours and mediated by cytokine release from sensitized T-cells.
- differentiation antigen:** A cell surface molecule expressed at a particular stage of development or on cells of a given lineage.
- DiGeorge syndrome:** Immunodeficiency caused by a congenital failure in thymic development resulting in a lack of mature functional T-cells.
- diversity (D) gene segments:** Found in the immunoglobulin heavy chain gene and T-cell receptor β and δ gene loci between the *V* and *J* gene segments. Encode part of the third hypervariable region in these antigen receptor chains.
- edema:** Swelling caused by accumulation of fluid in the tissues.
- effector cells:** Cells which carry out an immune function, e.g. cytokine release, cytotoxicity.
- ELISA (enzyme-linked immunosorbent assay):** Assay for detection or quantitation of an antibody or antigen using a ligand (e.g. an anti-immunoglobulin) conjugated to an enzyme which changes the color of a substrate.
- endocytosis:** Cellular ingestion of macromolecules by invagination of plasma membrane to produce an intracellular vesicle which encloses the ingested material.
- endogenous:** From within.
- endosomes:** Intracellular smooth surfaced vesicles in which endocytosed material passes on its way to the lysosomes.
- endotoxin:** Pathogenic cell wall-associated lipopolysaccharides of Gram-negative bacteria.
- eosinophil:** A class of granulocyte, the granules of which contain toxic cationic proteins.
- epitope:** That part of an antigen recognized by an antigen receptor (*see* antigenic determinant).
- Epstein–Barr virus (EBV):** The virus responsible for infectious mononucleosis and Burkitt's lymphoma. Used to immortalize human B-cells *in vitro*.
- equivalence:** The ratio of antibody to antigen at which immunoprecipitation of the reactants is virtually complete.
- erythema:** The redness produced during inflammation due to erythrocytes entering tissue spaces.
- erythropoiesis:** Erythrocyte production.
- exotoxin:** Pathogenic protein secreted by bacteria.
- exudate:** The extravascular fluid (containing proteins and cellular debris) which accumulates during inflammation.
- Fab:** Monovalent antigen-binding fragment obtained following papain digestion of immunoglobulin. Consists of an intact light chain and the N-terminal V_H and C_H1 domains of the heavy chain.
- F(ab)₂:** Bivalent antigen-binding fragment obtained following pepsin digestion of immunoglobulin. Consists of both light chains and the N-terminal part of both heavy chains linked by disulfide bonds.
- Fas:** A member of the TNF receptor gene family. Engagement of Fas (CD95) on the surface of the cell by the Fas ligand (CD178) present on cytotoxic cells, can trigger apoptosis in the Fas-bearing target cell.
- Fc:** Crystallizable, non-antigen binding fragment of an immunoglobulin molecule obtained following papain digestion. Consists of the C-terminal portion of both heavy chains which is responsible for binding to Fc receptors and C1q.
- Fc receptors:** Cell surface receptors which bind the Fc portion of particular immunoglobulin classes.
- fibroblast:** Connective tissue cell which produces collagen and plays an important part in wound healing.
- fluorescein isothiocyanate (FITC):** Green fluorescent dye used to 'tag' antibodies for use in immunofluorescence.
- fluorescent antibody:** An antibody conjugated to a fluorescent dye such as FITC.
- follicular dendritic cell:** MHC class II-negative Fc receptor-positive dendritic cells which bear immune complexes on their surface and are probably involved in the generation of antibody-secreting cells and maintenance of B-cell memory in germinal centres. (N.B. a different cell type to interdigitating dendritic cells).
- framework regions:** The relatively conserved amino acid sequences which flank the hypervariable regions in immunoglobulin and T-cell receptor variable regions and maintain a common overall structure for all V-region domains.
- Freund's adjuvant:** Complete Freund's adjuvant is an emulsion of aqueous antigen in mineral oil that contains heat-killed *Mycobacteria*. Incomplete Freund's adjuvant lacks the *Mycobacteria*.
- gamma globulin:** The serum proteins, mostly immunoglobulins, which have the greatest mobility towards the cathode during electrophoresis.
- germ line:** The arrangement of the genetic material as transmitted through the gametes.

germinal center: Discrete areas within lymph node and spleen where B-cell maturation and memory development occur.

giant cell: Large multinucleate cell derived from fused macrophages and often present in granulomas.

glomerulonephritis: Inflammation of renal glomerular capillary loops, often resulting from immune complex deposition.

graft versus host (g.v.h.) reaction: Reaction occurring when T lymphocytes present in a graft recognize and attack host cells.

granulocyte: Myeloid cells containing cytoplasmic granules (i.e. neutrophils, eosinophils and basophils).

granuloma: A tissue nodule containing proliferating lymphocytes, fibroblasts, and giant cells and epithelioid cells (both derived from activated macrophages), which forms due to inflammation in response to chronic infection or persistence of antigen in the tissues.

granzymes: Serine esterases present in the granules of cytotoxic T lymphocytes and NK cells. They induce apoptosis in the target cell which they enter through perforin channels inserted into the target cell membrane by the cytotoxic lymphocyte.

gut-associated lymphoid tissue (GALT): Includes Peyer's patches, appendix, and solitary lymphoid nodules in the submucosa.

H-2: The mouse major histocompatibility complex (MHC).

haplotype: The set of allelic variants present at a given genetic region.

hapten: A low molecular weight molecule that is recognized by preformed antibody but is not itself immunogenic unless conjugated to a 'carrier' molecule which provides epitopes recognized by helper T-cells.

helper T lymphocyte (Th): A subclass of T-cells which provide help (in the form of cytokines and/or cognate interactions) necessary for the expression of effector function by other cells in the immune system.

hemagglutinin: Any molecule which agglutinates erythrocytes.

hematopoiesis: The production of erythrocytes and leukocytes.

high endothelial venule (HEV): Capillary venule composed of specialized endothelial cells allowing migration of lymphocytes into lymphoid organs.

hinge region: Amino acids between the Fab and Fc regions of immunoglobulin which permit flexibility of the molecule.

histamine: Vasoactive amine present in basophil and mast cell granules which, following degranulation,

causes increased vascular permeability and smooth muscle contraction.

HLA (human leukocyte antigen): The human major histocompatibility complex (MHC).

humanized antibody: A genetically engineered monoclonal antibody of non-human origin in which all but the antigen-binding CDR sequences have been replaced with sequences derived from human antibodies. This procedure is carried out to minimize the immunogenicity of therapeutic monoclonal antibodies.

humoral: Pertaining to extracellular fluid such as plasma and lymph. The term humoral immunity is used to denote antibody-mediated immune responses.

hybridoma: Hybrid cell line obtained by fusing a lymphoid tumor cell with a lymphocyte which then has both the immortality of the tumor cell and the effector function (e.g. monoclonal antibody secretion) of the lymphocyte.

hypersensitivity: Excessive immune response which leads to undesirable consequences, e.g. tissue or organ damage.

hypervariable regions: Those amino acid sequences within the immunoglobulin and T-cell receptor variable regions which show the greatest variability and contribute most to the antigen or peptide-MHC binding site.

idiotope: An epitope made up of amino acids within the variable region of an antibody or T-cell receptor which reacts with an anti-idiotope.

idiotype: The complete set of idiotopes in the variable region of an antibody or T-cell receptor which react with an anti-idiotypic serum.

idiotype network: A regulatory network based on interactions of idiotypes and anti-idiotypes present on antibodies and T-cell receptors.

immune complex: Complex of antibody bound to antigen which may also contain complement components.

immunoabsorption: Method for removal of antibody or antigen by allowing it to bind to solid phase antigen or antibody.

immunofluorescence: Technique for detection of cell or tissue-associated antigens by the use of a fluorescently-tagged ligand (e.g. an anti-immunoglobulin conjugated to fluorescein isothiocyanate).

immunogen: Any substance which elicits an immune response. Whilst all immunogens are antigens, not all antigens are immunogens (*see* hapten).

immunoglobulin superfamily: Large family of proteins characterized by possession of 'immunoglobulin-type' domains of approximately 110 amino acids folded into two β -pleated sheets. Members include

- immunoglobulins, T-cell receptors and MHC molecules.
- immunological synapse:** A contact point between the T-cell and antigen-presenting cell which is generated by reorganization and clustering of cell surface molecules in lipid rafts. The synapse facilitates interactions between TCR and MHC and between adhesion molecules, thereby potentiating the TCR-mediated activation signal.
- inflammation:** The tissue response to trauma, characterized by increased blood flow and entry of leukocytes into the tissues, resulting in swelling, redness, elevated temperature and pain.
- innate immunity:** Immunity which is not intrinsically affected by prior contact with antigen, i.e. all aspects of immunity not directly mediated by lymphocytes.
- interdigitating dendritic cell:** MHC class II-positive, Fc receptor-negative, antigen-presenting dendritic cell found in T-cell areas of lymph nodes and spleen. (N.B. a different cell type to follicular dendritic cells).
- interferons (IFN):** IFN α is derived from various leukocytes, IFN β from fibroblasts and IFN γ from T lymphocytes. All three types induce an anti-viral state in cells and IFN γ acts as a cytokine in the regulation of immune responses.
- interleukins (IL):** Designation for some of the cytokines secreted by leukocytes.
- internal image:** An epitope on an anti-idiotypic which binds in a way that structurally and functionally mimics the antigen.
- invariant chain:** A polypeptide which binds MHC class II molecules in the endoplasmic reticulum, directs them to the late endosomal compartment and prevents premature association with self peptides.
- Ir (immune response) genes:** The genes, including those within the MHC, that together determine the overall level of immune response to a given antigen.
- isotype:** An antibody constant region structure present in all normal individuals, i.e. antibody class or subclass.
- ITAM:** Immunoreceptor Tyrosine-based Activation Motifs are consensus sequences for src-family tyrosine kinases. These motifs are found in the cytoplasmic domains of several signaling molecules including the signal transduction units of lymphocyte antigen receptors and of Fc receptors.
- ITIM:** Immunoreceptor Tyrosine-based Inhibitory Motifs present in the cytoplasmic domains of certain cell surface molecules, e.g. Fc γ RIIB, inhibitory NK cell receptors, and which mediate inhibitory signals.
- J chain:** A molecule which forms part of the structure of pentameric IgM and dimeric IgA.
- joining (J) gene segments:** Found in the immunoglobulin and T-cell receptor gene loci and, upon gene rearrangement, encode part of the third hyper-variable region of the antigen receptors.
- K (killer) cell:** Large granular lymphocyte which mediates antibody-dependent cellular cytotoxicity (ADCC), is Fc receptor positive, but does not rearrange or express either immunoglobulin or T-cell receptor genes.
- kinins:** A family of polypeptides released during inflammatory responses and which increase vascular permeability and smooth muscle contraction.
- KIRs:** Killer cell Immunoglobulin-like Receptors found on NK cells, some $\gamma\delta$ and some $\alpha\beta$ T-cells. KIRs recognize MHC class I molecules and, like the C-type lectin receptors also found on these cells, can either inhibit or activate the killer cells. If ITIM sequences are present in their cytoplasmic domain they are inhibitory. KIRs lacking ITIMs can associate with ITAM-containing adaptor molecules, in which case they can activate the killer cell.
- knockout:** The use of homologous genetic recombination in embryonal stem cells to replace a functional gene with a defective copy of the gene. The animals that are produced by this technique can be bred to homozygosity, thus allowing the generation of a null phenotype for that gene product.
- Kupffer cells:** Fixed tissue macrophages lining the blood sinuses in the liver.
- Langerhans' cell:** Fc receptor and MHC class II-positive antigen-presenting dendritic cell found in the skin.
- large granular lymphocyte (LGL):** Large lymphocytes which contain cytoplasmic granules and function as natural killer (NK) and killer (K) cells. Activated CD8 $^{+}$ cytotoxic T lymphocytes (Tc) also assume an LGL morphology.
- lectins:** A family of proteins, mostly of plant origin, which bind specific sugars on glycoproteins and glycolipids. Some lectins are mitogenic (e.g. PHA, ConA).
- leukotrienes:** Metabolic products of arachidonic acid which promote inflammatory processes (e.g. chemotaxis, increased vascular permeability) and are produced by a variety of cell types including mast cells, basophils and macrophages.
- ligand:** General term for a molecule recognized by a binding structure such as a receptor.
- linkage disequilibrium:** The occurrence of two alleles being inherited together at a greater frequency than that expected from the product of their individual frequencies.
- lipid raft:** Cholesterol- and glycosphingolipid-rich

membrane subdomain in which molecules involved in cellular activation become concentrated.

lipopolysaccharide (LPS): Endotoxin derived from Gram-negative bacterial cell walls which has inflammatory and mitogenic actions.

lymph: The tissue fluid which drains into and through the lymphatic system.

lymphadenopathy: Enlarged lymph nodes.

lymphokine: Cytokine produced by lymphocytes.

lymphokine-activated killer cells (LAK): Killer (K) and natural killer (NK) cells activated *in vitro* by IL-2 to give enhanced killing of target cells.

Lymphotoxin (also called TNF β): A T-cell derived cytokine which is cytotoxic for certain tumor cells and also has immunoregulatory functions.

lysosomes: Cytoplasmic granules containing hydrolytic enzymes involved in the digestion of phagocytosed material.

lysozyme: Anti-bacterial enzyme present in phagocytic cell granules, tears and saliva, which digests peptidoglycans in bacterial cell walls.

macrophage: Large phagocytic cell, derived from the blood monocyte, which also functions as an antigen-presenting cell and can mediate ADCC.

mannose binding protein: A member of the collectin family of calcium-dependent lectins, and an acute phase protein. It functions as a stimulator of the classical pathway of complement activation, and as an opsonin for phagocytosis by binding to mannose, a sugar residue usually found in an exposed form only on the surface of microorganisms.

marginal zone: The outer area of the splenic periarteriolar lymphoid sheath (PALS) which is rich in B cells, particularly those responding to thymus-independent antigens.

margination: Leukocyte adhesion to the endothelium of blood vessels in the early phase of an acute inflammatory reaction.

mast cell: A tissue cell with abundant granules which resembles the blood basophil. Both these cell types bear Fc receptors for IgE, which when crosslinked by IgE and antigen cause degranulation and the release of a number of mediators including histamine and leukotrienes.

medulla: Inner (central) region of an organ.

megakaryocyte: A bone marrow precursor of platelets.

membrane attack complex (MAC): Complex of complement components C5b–C9 which inserts as a pore into the membrane of target cells leading to cell lysis.

memory (immunological): A characteristic of the acquired immune response of lymphocytes whereby a second encounter with a given antigen produces a

secondary immune response; faster, greater and longer lasting than the primary immune response.

memory cells: Clonally expanded T- and B-cells produced during a primary immune response and which are 'primed' to mediate a secondary immune response to the original antigen.

MHC (major histocompatibility complex): A genetic region encoding molecules involved in antigen presentation to T-cells. Class I MHC molecules are present on virtually all nucleated cells and are encoded mainly by the H-2K, D, and L loci in mice and by HLA-A, B, and C in man, whilst class II MHC molecules are expressed on antigen-presenting cells (primarily macrophages, B-cells and interdigitating dendritic cells) and are encoded by H-2A and E in mice and HLA-DR, DQ, and DP in man. Allelic differences can be associated with the most intense graft rejection within a species.

MHC restriction: The necessity that T-cells recognize processed antigen only when presented by MHC molecules of the original haplotype associated with T-cell priming.

minor histocompatibility antigens: Non-MHC-encoded cell surface processed peptides which, in association with MHC-encoded molecules, contribute to graft rejection, albeit not usually as severe as that due to MHC mismatch.

mitogen: A substance which non-specifically induces lymphocyte proliferation.

mixed lymphocyte reaction (MLR): A T-cell proliferative response induced by cells expressing allogeneic MHC.

monoclonal antibody: Homogeneous antibody derived from a single B-cell clone and therefore all bearing identical antigen-binding sites and isotype.

monocyte: Mononuclear phagocyte found in blood and which is the precursor of the tissue macrophage.

mononuclear phagocyte system: A system comprising blood monocytes and tissue macrophages.

mucosal-associated lymphoid tissue (MALT): Lymphoid tissue present in the surface mucosa of the respiratory, gastrointestinal and genitourinary tracts.

multiple myeloma: Plasma cell malignancy resulting in high levels of monoclonal immunoglobulin in serum and of free light chains (Bence-Jones protein) in urine.

murine: Pertaining to mice.

myeloma protein: Monoclonal antibody secreted by myeloma cells.

negative selection: Deletion by apoptosis in the thymus of T-cells which recognize self peptides presented by self MHC molecules, thus preventing the development of autoimmune T-cells. Negative se-

- lection of developing B-cells is also thought to occur if they encounter high levels of self antigen in the bone marrow.
- neutrophil:** The major circulating phagocytic polymorphonuclear granulocyte. Enters tissues early in an inflammatory response and is also able to mediate antibody-dependent cellular cytotoxicity (ADCC).
- NK (natural killer) cell:** Large granular lymphocyte which does not rearrange nor express either immunoglobulin or T-cell receptor genes but is able to recognize and destroy certain tumor and virally-infected cells in an MHC and antibody-independent manner.
- NK-T cell:** NK1.1⁺ lymphoid cells with a morphology and granule content intermediate between T-cells and NK cells. They are potent producers of IL-4, may be CD4⁺8⁻ or CD4⁺8⁺, and express low levels of $\alpha\beta$ TCR with an invariant α chain and very restricted β chain specificity. Many of these TCR recognize antigens presented by the non-classical MHC-like molecule CD1. Their lectin-like NK1.1 receptor may recognize microbial carbohydrates.
- nude mouse:** Mouse which is T-cell deficient due to a homozygous gene defect (*nu/nu*) resulting in the absence of a thymus (and also lack of body hair).
- oncofetal antigen:** Antigen whose expression is normally restricted to the fetus but which may be expressed during malignancy in adults.
- opsonin:** Substance, e.g. antibody or C3b, which enhances phagocytosis by promoting adhesion of the antigen to the phagocyte.
- opsonization:** Coating of antigen with opsonin to enhance phagocytosis.
- PAF (platelet activating factor):** An alkyl phospholipid released by a variety of cell types including mast cells and basophils, which has immunoregulatory effects on lymphocytes and monocytes/macrophages as well as causing platelet aggregation and degranulation.
- paracortex:** The part of an organ (e.g. lymph node) which lies between the cortex and the medulla.
- pathogen-associated molecular pattern (PAMP):** Molecules such as lipopolysaccharide, peptidoglycan, lipoteichoic acids and mannans, which are widely expressed by microbial pathogens as repetitive motifs but are not present on host tissues. They are therefore utilized by the pattern recognition receptors (PRRs) of the immune system to distinguish pathogens from self antigens.
- pattern recognition receptor (PRR):** Receptors on professional antigen-presenting cells and phagocytes which enable them to recognize pathogen-associated molecular patterns (PAMPs). Amongst the large number of different PRRs are the mannose receptor (CD206) and the macrophage scavenger receptor (CD204).
- perforin:** Molecule produced by cytotoxic T-cells and NK cells which, like complement component C9, polymerizes to form a pore in the membrane of the target cell leading to cell death.
- periarteriolar lymphoid sheath (PALS):** The lymphoid tissue which forms the white pulp of the spleen.
- Peyer's patches:** Part of the gut associated lymphoid tissue (GALT) and found as distinct lymphoid nodules mainly in the small intestine.
- PHA (phytohemagglutinin):** A plant lectin which acts as a T-cell mitogen.
- phage antibody library:** A collection of cloned antibody variable region gene sequences which can be expressed as Fab or scFv fusion proteins with bacteriophage coat proteins. These can be displayed on the surface of the phages. The gene encoding a monoclonal recombinant antibody is enclosed in the phage particle and can be selected from the library by binding of the phage to specific antigen.
- phagocyte:** Cells, including monocytes/macrophages and neutrophils, which are specialized for the engulfment of cellular and particulate matter.
- phagolysosome:** Intracellular vacuole where killing and digestion of phagocytosed material occurs following the fusion of a phagosome with a lysosome.
- phagosome:** Intracellular vacuole produced following invagination of the cell membrane around phagocytosed material.
- phorbol myristate acetate (PMA):** A mitogenic phorbol ester which directly stimulates protein kinase C and acts as a tumor promoter.
- plaque forming cell (PFC):** Antibody-secreting plasma cell detected *in vitro* by its ability to produce a 'plaque' of lysed antigen-sensitized erythrocytes in the presence of complement.
- plasma cell:** Terminally differentiated B lymphocyte which actively secretes large amounts of antibody.
- pokeweed mitogen (PWM):** A plant lectin which is a T-cell dependent B-cell mitogen.
- polyclonal:** Many different clones, or the product of many different clones, e.g. polyclonal antiserum.
- poly-Ig receptor:** A receptor molecule which specifically binds J-chain containing polymeric Ig, i.e. dimeric secretory IgA and pentameric IgM, and transports it across mucosal epithelium.
- positive selection:** The selection of those developing T-cells in the thymus which are able to recognize self

- MHC molecules. This occurs by preventing apoptosis in these cells.
- precipitin:** Precipitate of antibody and multivalent antigen due to the formation of high molecular weight complexes.
- primary immune response:** The relatively weak immune response which occurs upon the first encounter of naive lymphocytes with a given antigen.
- primary lymphoid organs:** The sites at which immunocompetent lymphocytes develop, i.e. bone marrow and thymus in mammals.
- prime:** The process of giving an initial sensitization to antigen.
- prostaglandins:** Acidic lipids derived from arachidonic acid which are able to increase vascular permeability, mediate fever, and can both stimulate and inhibit immunological responses.
- proteasome:** Cytoplasmic proteolytic enzyme complex involved in antigen processing for association with MHC.
- protein A:** *Staphylococcus aureus* cell wall protein which binds to the Fc region of IgG.
- protein tyrosine kinases:** Enzymes which are able to phosphorylate proteins on tyrosines, and often act in a cascade-like fashion in the signal transduction systems of cells.
- prozone effect:** The loss of immune precipitation or agglutination which occurs when antibody concentration is increased to an extent that the antibody is in such excess that it is no longer able to effectively cross-link the antigen. A similar phenomenon may occur in antigen excess.
- Qa antigens:** 'Non-classical' MHC class I molecules of mice.
- recombination signal sequence (RSS):** Conserved heptamer (7-nucleotide)-nonamer (9-nucleotide) sequences, separated by a 12 or 23 base spacer, which occur 3' of variable gene segments, 5' and 3' of diversity gene segments, and 5' of joining gene segments, in both immunoglobulin and T cell receptor genes. They function as recognition sequences for the recombinase enzymes that mediate the gene rearrangement process involved in the generation of lymphocyte antigen receptor diversity.
- respiratory burst:** The increased oxidative metabolism which occurs in phagocytic cells following activation.
- reticuloendothelial system (RES):** A rather old term for the network of phagocytes and endothelial cells throughout the body.
- rheumatoid factor:** IgM, IgG and IgA autoantibodies to IgG, particularly the Fc region.
- rosette:** Particles or cells bound to the surface of a lymphocyte (e.g. sheep erythrocytes around a human T-cell).
- scFv:** A single chain molecule composed of the variable regions of an antibody heavy and light chain joined together by a flexible linker.
- SCID (severe combined immunodeficiency):** Immunodeficiency affecting both T and B lymphocytes.
- secondary immune response:** The qualitatively and quantitatively improved immune response which occurs upon the second encounter of primed lymphocytes with a given antigen.
- secretory component:** Proteolytic cleavage product of the poly-Ig receptor which remains associated with dimeric IgA in sero-mucus secretions.
- secretory IgA:** Dimeric IgA found in sero-mucus secretions.
- somatic hypermutation:** The enhanced rate of point mutation in the immunoglobulin variable region genes which occurs following antigenic stimulation and acts as a mechanism for increasing antibody diversity and affinity.
- stem cell:** Multipotential cell from which differentiated cells derive.
- superantigen:** An antigen which reacts with all the T-cells belonging to a particular T-cell receptor V region family, and which therefore stimulates (or deletes) a much larger number of cells than does conventional antigen.
- surface plasmon resonance:** A technique based upon changes in the angle of reflected light which occur upon ligand binding to an immobilized target molecule on a biosensor chip. This permits the observation of protein-protein interactions (such as antibody binding to an antigen) in 'real-time', i.e. by continuous monitoring of the association and dissociation of the reversible reaction.
- switch sequences:** Highly conserved repetitive sequences which mediate class switching in the immunoglobulin heavy chain gene locus.
- syngeneic:** Genetically identical, e.g. a fully inbred strain of mice.
- TAP:** The Transporters associated with Antigen Processing (TAP-1 and TAP-2) are molecules which carry antigenic peptides from the cytoplasm into the lumen of the endoplasmic reticulum for incorporation into MHC class I molecules.
- T-cell receptor (TCR):** The heterodimeric antigen receptor of the T lymphocyte exists in two alternative forms, consisting of α and β chains, or γ and δ chains. The $\alpha\beta$ TCR recognizes peptide fragments of protein antigens presented by MHC molecules on cell surfaces. The function of the $\gamma\delta$ TCR is less clearly de-

fined but it can recognize native proteins on the cell surface.

T-dependent antigen: An antigen which requires helper T-cells in order to elicit an antibody response.

T-independent antigen: An antigen which is able to elicit an antibody response in the absence of T-cells.

thymocyte: Developing T-cell in the thymus.

titer: Measure of the relative 'strength' (a combination of amount and avidity) of an antibody or anti-serum, usually given as the highest dilution which is still operationally detectable in, for example, an ELISA.

tolerance: Specific immunological unresponsiveness.

tolerogen: An antigen used to induce tolerance. Often depends more on the circumstances of administration (e.g. route and concentration) than on any inherent property of the molecule.

toxoid: Chemically or physically modified toxin that is no longer harmful but retains immunogenicity.

tumor necrosis factor (TNF, also called TNF α): Together with the related cytokine lymphotoxin (TNF β), was originally named for its cytotoxic effect on certain tumor cells, but also has immunoregulatory functions.

variable (V) gene segments: Genes that rearrange together with *D* (diversity) and *J* (joining) gene segments in order to encode the variable region amino acid sequences of immunoglobulins and T-cell receptors.

vasoactive amines: Substances including histamine and 5-hydroxytryptamine which increase vascular permeability and smooth muscle contraction.

xenogeneic: Genetic differences between species.

xenograft: A tissue or organ graft between individuals of different species.

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